

MD. ASHHAB SADIQ

sadiq.kuet.urp@gmail.com | +880 1794200784 | [Portfolio](#) | [LinkedIn](#) | Dhaka-1219, Bangladesh

RESEARCH PROFILE

Research background in geospatial analysis, remote sensing, environmental assessment, and spatial modeling. Research experience includes using satellite observations, GIS, and machine learning to study climate and environmental processes, with studies on meteorological drought, groundwater vulnerability, and human activity-related CO₂ emissions.

EDUCATION

Master of Science in GIS for Environment and Development (MSGED)

Jahangirnagar University (JU), Dhaka-1342, Bangladesh

Total Credits: 42 | Expected Graduation: June 2027

Relevant Coursework (In Progress): Principles of Environment, Fundamentals of GIS, GIS in Flood Plain and Watershed Management, GIS in Climate Change and Sustainable Development, Research Techniques

Bachelor of Urban and Regional Planning (BURP)

Khulna University of Engineering & Technology (KUET), Khulna-9203, Bangladesh

Total Credits: 162 | CGPA: **3.59** out of 4.00 | Graduation: April 2023

Achievements: Dean's Award (2020, 2021) and University Merit Scholarship (2019–2021)

Relevant Coursework: GIS and Remote Sensing, Environmental Planning and Management, Water Resources Planning, Statistics I & II, Hazards and Disaster Management, Research Methodology

PUBLICATIONS

- Raisa, S. S., Sarkar, S. K., & **Sadiq, M. A.** (2024). Advancing groundwater vulnerability assessment in Bangladesh: A comprehensive machine learning approach. **Groundwater for Sustainable Development**, 101128. [DOI \(Q1\)](#)
- Sadiq, M. A.**, Sarkar, S. K., & Raisa, S. S. (2023). Meteorological drought assessment in northern Bangladesh: A machine learning-based approach considering remote sensing indices. **Ecological Indicators**, 157, 111233. [DOI \(Q1\)](#)
- Haque, M. N., Abtahee, M., Islam, A. A., & **Sadiq, M. A.** (2023). Students' perception of environmental sustainability exercises at higher education institutions in Bangladesh. **Frontiers in Engineering and Built Environment**. [DOI \(Q2\)](#)

RESEARCH EXPERIENCE

Meteorological Drought Assessment in Northern Bangladesh

Undergraduate Thesis Research | KUET

- Developed a remote sensing and machine learning-based framework to assess meteorological drought across northern Bangladesh from 2010–2019.
- Generated drought indicators using Google Earth Engine, calculated multi-timescale SPI in R, and applied Random Forest modeling in Python to evaluate the relative importance of drought factors.
- Produced drought distribution maps and found surface-related indices showed greater relative importance for short-term drought, whereas vegetation-related indices were more important for long-term drought.

Groundwater Vulnerability (GWV) Assessment in Bangladesh

Research Collaboration | KUET

- Developed a multi-criteria GWV assessment framework integrating GIS and Random Forest modeling.
- Integrated 21 factors across 4 categories: topography, meteorology, socio-economic, and land use & geology.
- Produced 4 categorical GWV maps and 1 final composite GWV map of Bangladesh, with land use & geological factors emerging as the dominant contributors (50.82%), followed by topographic factors (22.69%).

Assessment of the Impact of MRT-6 on Reducing the Carbon Footprint of Dhaka City

Social Science Research Council-funded Project, Ministry of Planning, Bangladesh | Manuscript Under Review

- Analyzed commuter modal shifts and applied mode-specific emission factors and MRT-6 electricity consumption to estimate gross and net CO₂ emission reductions.
- Estimated 26,767 tons of net annual CO₂ savings from MRT-6, while identifying first-mile and last-mile connectivity as a key factor limiting its potential carbon reduction benefits.

Students' Perception of Environmental Sustainability at Higher Education Institutions in Bangladesh

Research Collaboration | KUET

- Analyzed sustainability perceptions using different statistical tests and models, identifying significant relationships between demographic characteristics and sustainability awareness, attitudes, and participation.

PROFESSIONAL EXPERIENCE

Research Consultant

June 2026 - Present

Center for Environmental and Geographic Information Services (CEGIS)

- Prepare and revise technical reports for multidisciplinary planning and consulting projects, incorporating expert and client feedback to strengthen analyses and finalize project deliverables.
- Contribute to technical proposals for environment and development related projects by integrating sector-specific technical approaches and coordinating inputs from specialists across multiple sectors.

Junior GIS Analyst

August 2024 - December 2025

ACE Consultants Ltd. (A Subsidiary of SMEC International Pty. Ltd., Australia)

- Developed geospatial databases and land acquisition datasets for highway feasibility and design studies, including 22 road sections covering 1,480 km and the Dhaka–Chattogram National Highway (N1).
- Performed GIS quality control and georeferencing, and produced land acquisition, environmental, and design maps using satellite and drone imagery to support highway planning and design.

Assistant Urban Planner

July 2023 - August 2024

Sheltech Consultants (Pvt.) Ltd.

- Coordinated project initiation, inception, and survey activities for master planning across three sub-districts, supervising multiple field and GIS teams and managing community consultations.
- Prepared and managed spatial databases and GIS outputs, integrating socio-economic, environmental, and land-use survey data into planning analyses and technical reports.

TECHNICAL AND RESEARCH SKILLS

GIS: ArcGIS Pro, ArcMap, QGIS, Global Mapper, ArcGIS Online; Geodatabase Development, Spatial Analysis, Mapping

Remote Sensing & Earth Observation: Google Earth Engine; Satellite Data Analysis, Environmental Change Detection

Geospatial Programming: Python, R; Libraries or Modules including ArcPy, Pandas, NumPy, Scikit-learn

Machine Learning & Spatial Modeling: Random Forest, XGBoost, Moran's I; Time-series Analysis, Spatial Correlation

Survey & Statistical Analysis: SPSS, Microsoft Excel, KoboToolbox, Google Forms; Survey Design, Data Collection

Reporting & Visualization: Microsoft Word, Microsoft PowerPoint, ArcGIS StoryMaps, Canva

RELEVANT ACADEMIC PROJECTS

Regional Diversity Assessment of Bangladesh: Applied GIS and remote sensing techniques to derive LULC, LST, NDVI, NDWI, NDMI, and integrate socio-economic, structural, and ecological factors to assess spatial variation among districts.

Spatiotemporal Assessment of Service Facilities in Khulna City: Evaluated land-use change, waterlogging, drainage, and service conditions using GIS analysis, stakeholder assessment, and questionnaire surveys.

Geospatial Analysis of Urban and Environmental Conditions in Rajshahi City: Applied GIS and remote sensing techniques to analyze service accessibility, land use, flood hazards, and hydrological features across Rajshahi City.

CERTIFICATIONS

Verifiable at [LinkedIn](#)

- Geographic Information System (GIS) Specialization: University of California (via Coursera)
- Data Analysis with R Programming: Google (via Coursera)
- Cities and Climate Change: United Nations Institute for Training and Research (via UNITAR)
- Cartography and The Location Advantage: ESRI
- Python Data Structures: University of Michigan (via Coursera)

EXTRACURRICULAR ACTIVITIES

Assistant Research and Publication Secretary (URP Association - KUET)

November 2021 - July 2022

- Organized academic events for the Department of Urban and Regional Planning, including a workshop on “Scientific Research Article Writing” and other student-focused initiatives.

Head of Event Management (Design Integrated Society of KUET - DISK)

August 2019 - August 2021

- Involved in organizing two national events at the campus, “Plannation” and “Plannation 2.0”.

REFEREES

Tanmoy Chakraborty

Assistant Professor
Department of Urban and Regional
Planning, KUET, Khulna: 9203, Bangladesh
Email: tanmoy.chakraborty@urp.kuet.ac.bd

Showmitra Kumar Sarkar

Assistant Professor
Department of Urban and Regional
Planning, KUET, Khulna: 9203, Bangladesh
Email: showmitrasarkar@urp.kuet.ac.bd

Khondaker Md Mohiuddin Ekram

Assistant Professor
Department of Urban and Regional
Planning, KUET, Khulna: 9203, Bangladesh
Email: ekram@g.harvard.edu